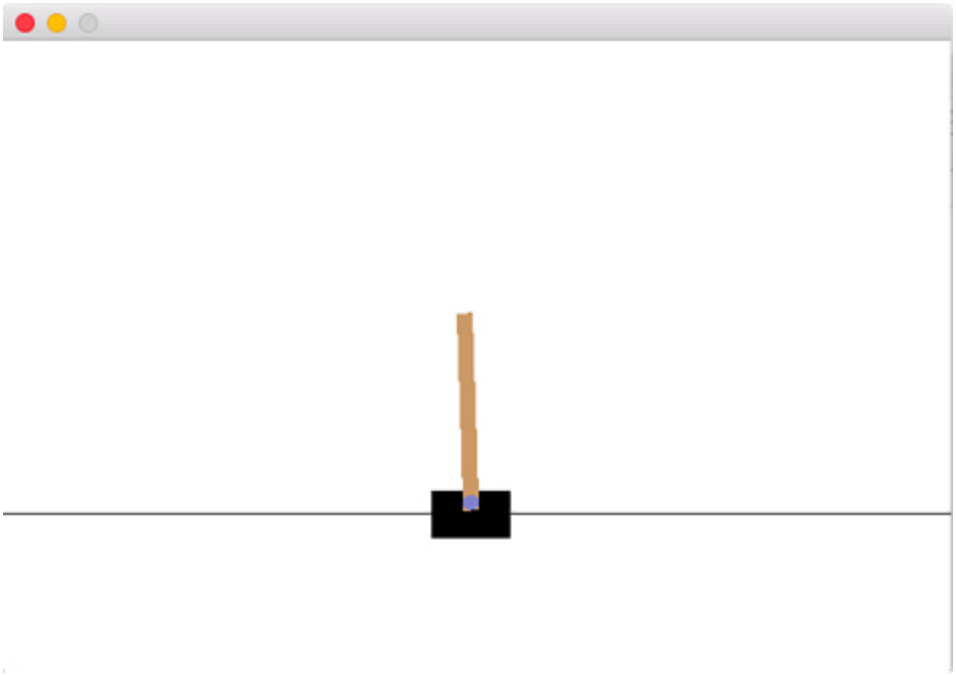
Listing 4.1. Listing the OpenAI Gym environments

```
from gym import envs
envs.registry.all()
```

There are hundreds of environments to choose from (797 in v0.9.6). Unfortunately, some of these environments require licenses (MuJoCo) or external dependencies (Box2D, Atari) and will therefore require a bit of setup time. We will be starting with a simple example, CartPole (figure 4.9), to avoid any unnecessary complications and to get us coding right away.

Figure 4.9. A screenshot from the CartPole game environment in OpenAI Gym. There is a cart that can roll left or right, and on top of it is a pole on a pivot. The goal is to balance the pole upright on the cart by carefully moving the cart left or right.



4.3.1. CartPole

The CartPole environment falls under OpenAI's Classic Control section, and it has a very simple objective—don't let the pole fall over. It's the game equivalent of trying to balance a pencil on the tip of your finger. In order to balance the pole successfully, you have to apply just the right amount of small left and right movements to the cart. In this environment, there are only two actions that correspond to making a small push left or right.